DOĞU ERBAŞ

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METU Computer Engineering student with experience in C/C++, C#, backend development, computer graphics, robotics software, and embedded systems coursework. Interested in game development, low-level systems, robotics, and simulation-based applications.

EDUCATION

Middle East Technical University (METU) — Ankara, Turkey
B.S. in Computer Engineering — Expected Graduation: June 2027

EXPERIENCE

METU Robotics and AI Technologies Research Center — June 2025 – June 2026
Researcher

- Contributed to C++ software components and simulation workflows for a Lunar Rover prototype using the MuJoCo physics engine.
- Worked as part of a research team on robotics software pipelines, real-time simulation, experimental testing, and data logging scripts.
- Supported OptiTrack-based tracking environments for navigation, spatial analysis, and rover testing workflows.

Europower Energy and Automation Technologies Inc. — August 2025 – September 2025
Software Intern

- Developed two internal applications using C#, .NET MAUI, ASP.NET Core, and SQL Server.
- Built backend APIs, database operations, and mobile interfaces for factory equipment tracking and transformer QC data management systems.
- Collaborated with engineering teams to test and refine applications in real-world production conditions.

PROJECTS

Ray Tracer Implementation (C++)

- Implemented recursive ray casting with Blinn-Phong shading, shadow rays, mirror reflection, and geometric intersections.
- Worked with camera transformations, linear algebra, scene representation, and physically inspired rendering concepts.

Forward Rendering Pipeline (C++)

- Engineered a software-based rasterization pipeline with modeling, viewing, projection, culling, and clipping transformations.
- Implemented a complete transformation pipeline from object-space geometry to screen-space rendering.

Three-Port EV Charging Cabinet Controller (C, PIC18F8722, EUSART, ADC, Timers)

- Contributed to firmware for a three-port EV charging cabinet controller using EUSART communication, ADC thermal sensing, and timer-driven control logic.
- Implemented state-machine behavior, serial command handling, current-limit logic, and 7-segment display interaction in C.

Transformer QC Data Management System (C#, .NET MAUI, ASP.NET Core, SQL Server)

- Built a dual-database application to automate transformer test record fetching, verification, and reporting.

Factory Equipment Tracking System (C#, .NET MAUI, ASP.NET Core, SQL Server)

- Developed a mobile asset management tool using MVVM architecture for real-time tracking of facility equipment.

TECHNICAL SKILLS & LANGUAGES

Languages: C/C++, C#, Python, SQL, HTML/CSS, Bash, Assembly
Tools & Platforms: Linux, Git, .NET MAUI, ASP.NET Core, SQL Server, OpenGL
Interests: Computer Graphics, Low-Level Systems, Embedded Systems, Machine Learning, Game Development
Foreign Languages: English (Advanced, C1/C2), German (Intermediate, A2/B1)

EXTRACURRICULAR

Competitive Swimmer. — Mountaineering and Technical Climbing enthusiast. — Analog & Digital Photographer.